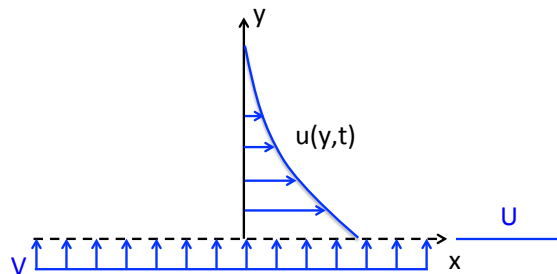


Problem Set #3

Due Fri., Apr. 24 by 11:59pm

Please hand in your completed problem sets as PDF files to the course Canvas site. Working together with others is fine (and encouraged), but you must write up each solution yourself.

1. Consider the unsteady two-dimensional flow of incompressible viscous fluid above a flat, porous plate of infinite extent. The plate lies along the x -axis and is initially at rest. At time $t = 0$, the plate suddenly starts moving to the right at velocity U . At the same time, fluid starts to be injected uniformly through the plate with velocity V .



- a) Find an equation that governs the x -component of velocity $u(y, t)$ of the fluid above the plate.
 - b) If the injection velocity V is constant in time (and non-zero), show that no similarity solution can exist.
 - c) Find a time-varying injection velocity $V(t)$ that will admit a similarity solution $f(\eta)$ for this problem. In this case, reduce the PDE of part (a) to an ODE for $f(\eta)$.
2. Use a shooting method to numerically solve the Blasius equation, $f''' + \frac{1}{2}ff'' = 0$. The attached spreadsheet sets up this calculation for you but needs to be completed. First, in cell D3, fill in a formula for f'' in terms of values at the previous η location. Then, copy line 3 to subsequent lines to integrate the Blasius equation. Continue copying until reaching $\eta = 10$.
- a) Note that cell D2 contains a guess for the wall shear stress, $f''(0) = 0.2$. Using this guess and $\Delta\eta = 0.1$, what is $f'(10)$? What value should f' have at this η location?
 - b) Now adjust the wall shear stress, keeping $\Delta\eta = 0.1$, until the $f'(10)$ converges to its correct freestream value. What is value of $f''(0)$ in this case?
 - c) The Blasius wall shear stress is known to be $f''(0) = 0.332$, which can be found by decreasing $\Delta\eta$ and taking more steps. Repeat part (b) with $\Delta\eta = 0.01$. What is value of $f''(0)$ in this case?
3. Consider a third-order polynomial as a model for the velocity profile in a boundary layer flow over a flat plate (with zero pressure gradient):

$$\frac{u(y)}{U_\infty} = a_0 + a_1\eta + a_2\eta^2 + a_3\eta^3, \quad \text{where } \eta = y/\delta.$$

In addition to the three boundary conditions used to construct the parabolic profile in class ($u(0) = 0$, $u(\delta) = U_\infty$, $\frac{\partial u}{\partial y}\big|_\delta = 0$), one more boundary condition is needed to construct the third-order profile. Two possible such boundary conditions are: “zero curvature at the wall” and “zero curvature at $y = \delta$.” The aim of this problem is to compare the performance of the profiles resulting from these two boundary conditions, relative to one another as well as relative to the parabolic profile discussed in lecture and the exact (Blasius) solution.

- a) Find the coefficients a_0, a_1, a_2, a_3 for the third-order profiles corresponding to the two different boundary conditions.
- b) On one figure, plot profiles corresponding to the exact Blasius solution (use your results of problem 2), the parabolic profile, and the two third-order profiles.
- c) Using the Momentum Integral Equation, evaluate the boundary layer thicknesses δ , δ^* , and θ and the wall shear stress τ_w as functions of x , for the two third-order profiles and compare to expressions obtained for the exact and parabolic profiles.
- d) Do either of the third order profiles offer any improvement over the parabolic profile? Briefly explain your findings.